```
clc;
clear;
close all;
```

INPUT PARAMETERS

```
b = [0.0048 0.0193 0.0289 0.0193 0.0048];
a = [1 -2.3695 2.3140 -1.0547 0.1874];

Bc = 8;                 % Coefficient word length
Bp = 16;                % Product word length
N = 200;                % Number of samples

x = [1 zeros(1,N-1)];   % Unit impulse
n = 0:N-1;
```

1. COEFFICIENT QUANTIZATION

```
S = 2^(Bc-1)-1;

bq = round(b*S)/S;
aq = round(a*S)/S;

% Keep the denominator leading coefficient normalized
bq = bq/aq(1);
aq = aq/aq(1);

CoeffError = [b-bq, a-aq];
```

2. IDEAL AND COEFFICIENT-QUANTIZED RESPONSE

```
y_ideal = filter(b,a,x);
y_coeff = filter(bq,aq,x);

e_coeff = y_ideal-y_coeff;

MaxCoeffError = max(abs(e_coeff));

signalEnergy = sum(y_ideal.^2);
errorEnergy = sum(e_coeff.^2);

if errorEnergy > 0
    SQNR_coeff = 10*log10(signalEnergy/errorEnergy);
else
    SQNR_coeff = Inf;
end
```

3. PRODUCT QUANTIZATION

```
Sp = 2^(Bp-1)-1;

y_prod = zeros(1,N);
v = zeros(1,4);

for k = 1:N

    % Feedforward products
    p = bq*x(k);

    % Feedback products
    fb = -aq(2:5).*v;

    % Quantize individual products
    p = round(p*Sp)/Sp;
    fb = round(fb*Sp)/Sp;

    % Sum the quantized products
    y_prod(k) = sum(p) + sum(fb);

    % Update delay states
    v = [y_prod(k), v(1:3)];

end

e_prod = y_coeff-y_prod;

MaxProdError = max(abs(e_prod));

signalEnergyProd = sum(y_coeff.^2);
errorEnergyProd = sum(e_prod.^2);

if errorEnergyProd > 0
    SQNR_prod = 10*log10(signalEnergyProd/errorEnergyProd);
else
    SQNR_prod = Inf;
end
```

4. ZERO-INPUT LIMIT-CYCLE TEST

```
x0 = zeros(1,N);

state = [0.01 0.01 0.01 0.01];
y_zero = zeros(1,N);

for k = 1:N

    acc = -aq(2:5)*state.';

    % Quantize output
```

```
    y_zero(k) = round(acc*Sp)/Sp;

    % Update states
    state = [y_zero(k), state(1:3)];

end

LimitAmplitude = max(abs(y_zero));
LimitSamples = find(abs(y_zero)>1e-12);
```

5. OVERFLOW / SATURATION TEST

```
xo = 1.2*ones(1,N);

y_over = filter(bq,aq,xo);

% Saturation to normalized range [-1,1]
y_sat = min(max(y_over,-1),1);

OverflowCount = sum(abs(y_over)>1);
MaxOverflow = max(abs(y_over));
```

6. DISPLAY RESULTS

```
fprintf('\n=============================================\n');
fprintf('    FINITE WORD-LENGTH EFFECTS IN IIR FILTERS\n');
fprintf('=============================================\n');

fprintf('Coefficient Word Length = %d bits\n',Bc);
fprintf('Product Word Length = %d bits\n',Bp);
fprintf('Number of Samples = %d\n',N);

fprintf('\n--- QUANTIZATION RESULTS ---\n');

fprintf('Maximum Coefficient-Quantized Output Error = %.6e\n', ...
    MaxCoeffError);
fprintf('Coefficient Quantization SQNR = %.2f dB\n', ...
    SQNR_coeff);

fprintf('Maximum Product Quantization Error = %.6e\n', ...
    MaxProdError);
fprintf('Product Quantization SQNR = %.2f dB\n', ...
    SQNR_prod);

fprintf('\n--- ZERO-INPUT LIMIT-CYCLE TEST ---\n');

fprintf('Maximum Zero-Input Amplitude = %.6e\n', ...
    LimitAmplitude);
fprintf('Number of Nonzero Samples = %d\n', ...
    length(LimitSamples));

fprintf('\n--- OVERFLOW / SATURATION TEST ---\n');
```

```
fprintf('Number of Overflow Samples = %d\n', ...
    OverflowCount);
fprintf('Maximum Unsaturated Output Magnitude = %.6f\n', ...
    MaxOverflow);


==========================================
    FINITE WORD-LENGTH EFFECTS IN IIR FILTERS
==========================================
Coefficient Word Length = 8 bits
Product Word Length = 16 bits
Number of Samples = 200

--- QUANTIZATION RESULTS ---
Maximum Coefficient-Quantized Output Error = 4.405764e-03
Coefficient Quantization SQNR = 32.23 dB
Maximum Product Quantization Error = 1.651951e-01
Product Quantization SQNR = 1.57 dB

--- ZERO-INPUT LIMIT-CYCLE TEST ---
Maximum Zero-Input Amplitude = 9.216590e-03
Number of Nonzero Samples = 177

--- OVERFLOW / SATURATION TEST ---
Number of Overflow Samples = 194
Maximum Unsaturated Output Magnitude = 1.359910
```

7. OUTPUT PARAMETER TABLE

```
Test = [
    "Coefficient Quantization"
    "Product Quantization"
    "Zero-Input Test"
    "Overflow Test"
];

MaximumValue = [
    MaxCoeffError
    MaxProdError
    LimitAmplitude
    MaxOverflow
];

SQNR_dB = [
    SQNR_coeff
    SQNR_prod
    NaN
    NaN
];

T = table(Test,MaximumValue,SQNR_dB);
```

```
fprintf('\n--- OUTPUT PARAMETER TABLE ---\n');
disp(T);
```

```
--- OUTPUT PARAMETER TABLE ---
                Test                  MaximumValue    SQNR_dB
    ______________________________    ____________    _______

    "Coefficient Quantization"          0.0044058      32.233
    "Product Quantization"                 0.1652      1.5673
    "Zero-Input Test"                   0.0092166         NaN
    "Overflow Test"                        1.3599         NaN
```

8. DISPLAY QUANTIZED COEFFICIENTS

```
fprintf('\nOriginal Numerator Coefficients:\n');
disp(b);

fprintf('Quantized Numerator Coefficients:\n');
disp(bq);

fprintf('Original Denominator Coefficients:\n');
disp(a);

fprintf('Quantized Denominator Coefficients:\n');
disp(aq);

fprintf('Coefficient Quantization Errors:\n');
disp(CoeffError);
```

```
Original Numerator Coefficients:
    0.0048    0.0193    0.0289    0.0193    0.0048

Quantized Numerator Coefficients:
    0.0079    0.0157    0.0315    0.0157    0.0079

Original Denominator Coefficients:
    1.0000   -2.3695    2.3140   -1.0547    0.1874

Quantized Denominator Coefficients:
    1.0000   -2.3701    2.3150   -1.0551    0.1890

Coefficient Quantization Errors:
  Columns 1 through 7

   -0.0031    0.0036   -0.0026    0.0036   -0.0031         0    0.0006

  Columns 8 through 10

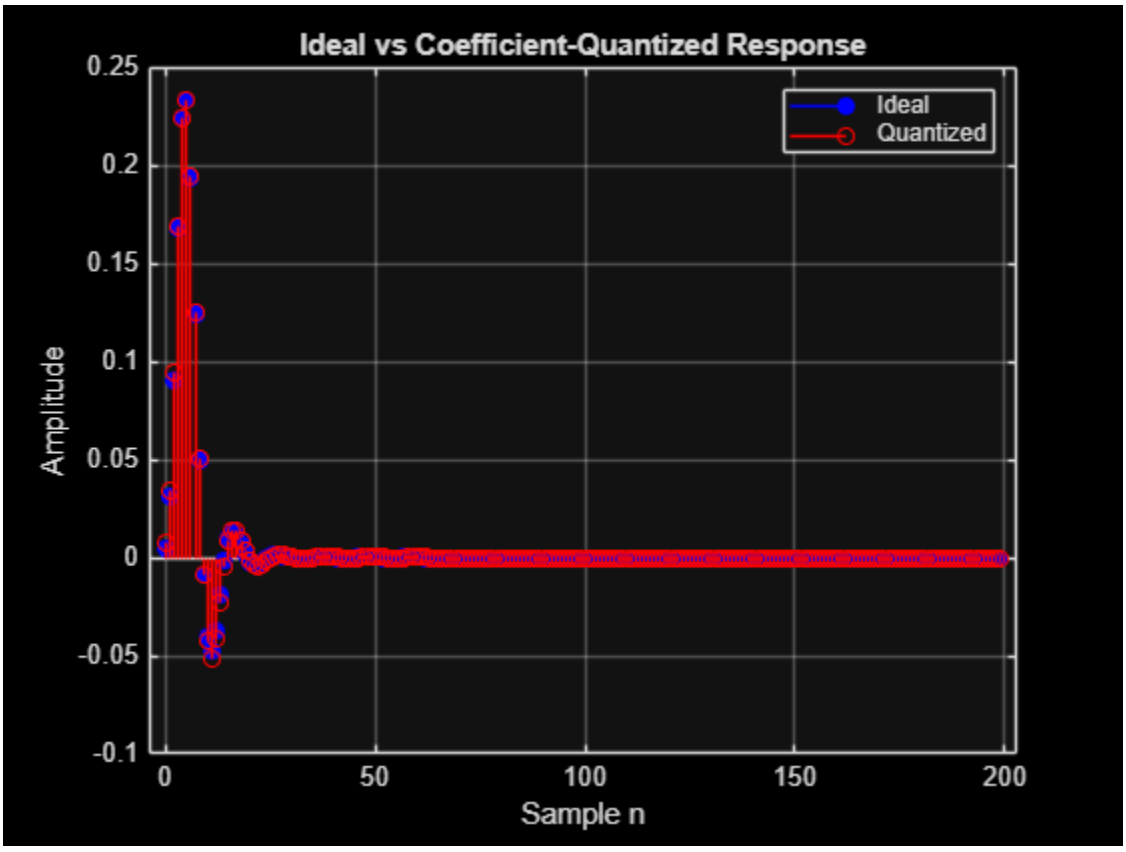
   -0.0010    0.0004   -0.0016
```

9. GRAPH 1: IDEAL VS QUANTIZED RESPONSE

```
figure('Name','Coefficient Quantization');

stem(n,y_ideal,'b','filled');
hold on;
stem(n,y_coeff,'r');

grid on;
xlabel('Sample n');
ylabel('Amplitude');
title('Ideal vs Coefficient-Quantized Response');
legend('Ideal','Quantized');
```



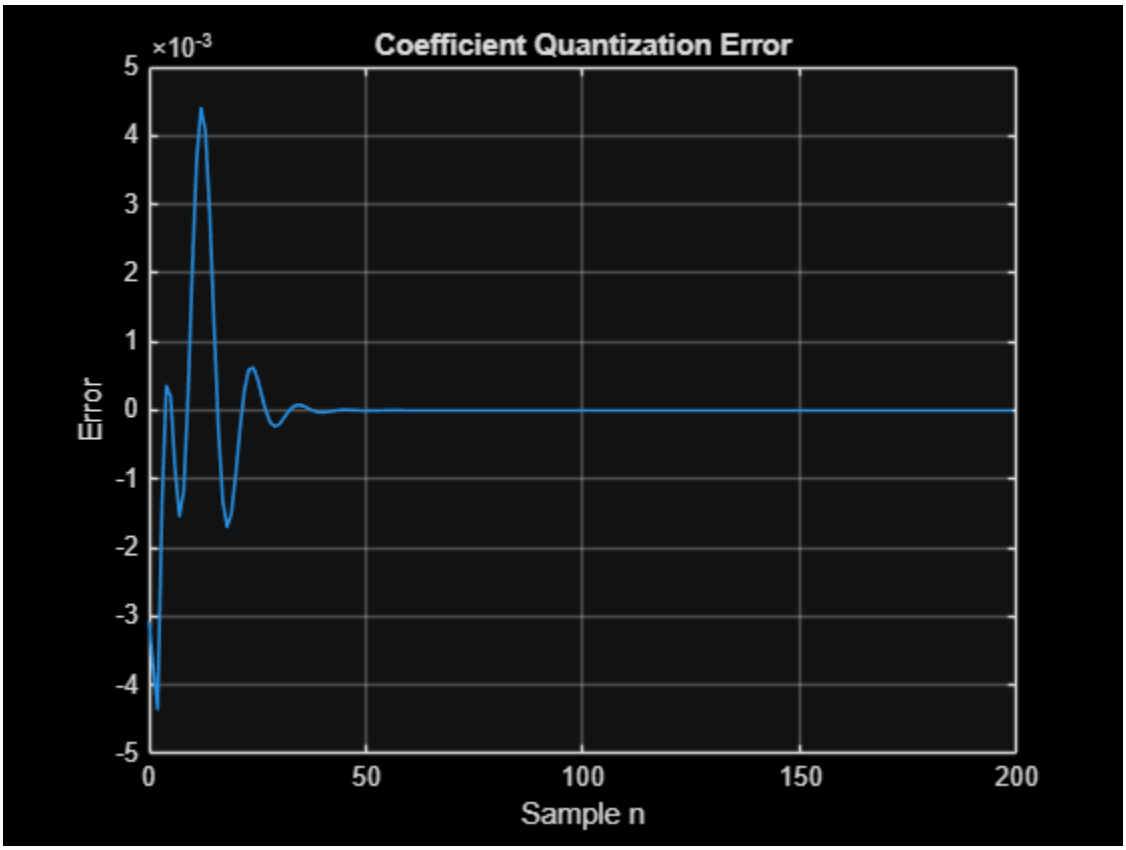
10. GRAPH 2: COEFFICIENT QUANTIZATION ERROR

```
figure('Name','Coefficient Quantization Error');

plot(n,e_coeff,'LineWidth',1.2);

grid on;
```

```
xlabel('Sample n');
ylabel('Error');
title('Coefficient Quantization Error');
```

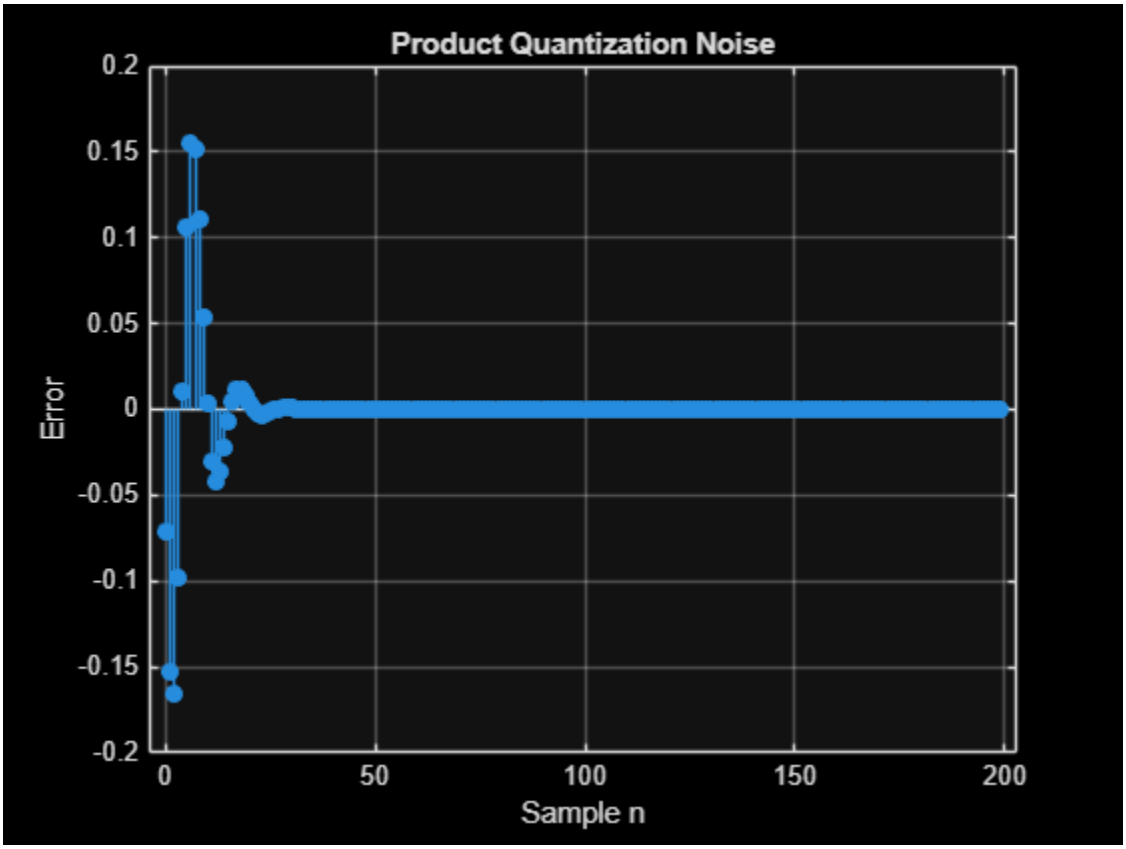


11. GRAPH 3: PRODUCT QUANTIZATION ER-ROR

```
figure('Name','Product Quantization Error');

stem(n,e_prod,'filled');

grid on;
xlabel('Sample n');
ylabel('Error');
title('Product Quantization Noise');
```

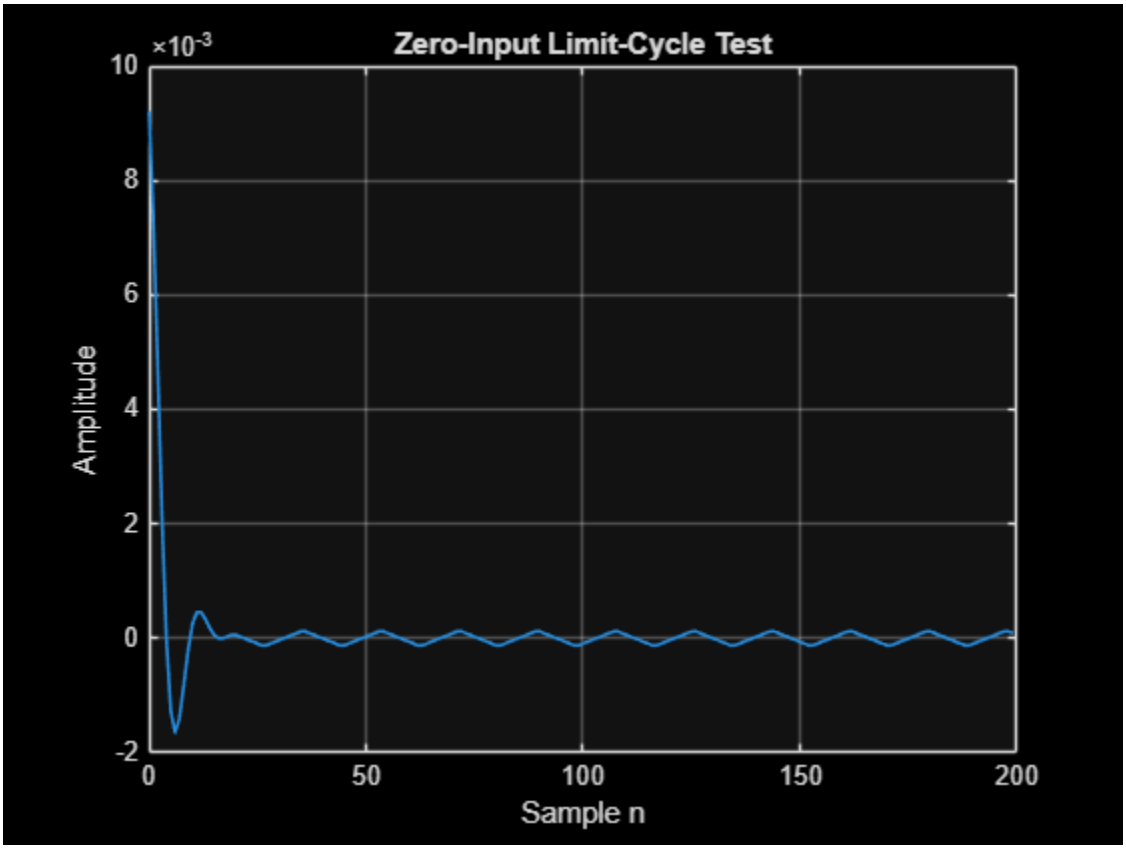


12. GRAPH 4: ZERO-INPUT LIMIT-CYCLE TEST

```
figure('Name','Zero-Input Limit-Cycle Test');

plot(n,y_zero,'LineWidth',1.2);

grid on;
xlabel('Sample n');
ylabel('Amplitude');
title('Zero-Input Limit-Cycle Test');
```



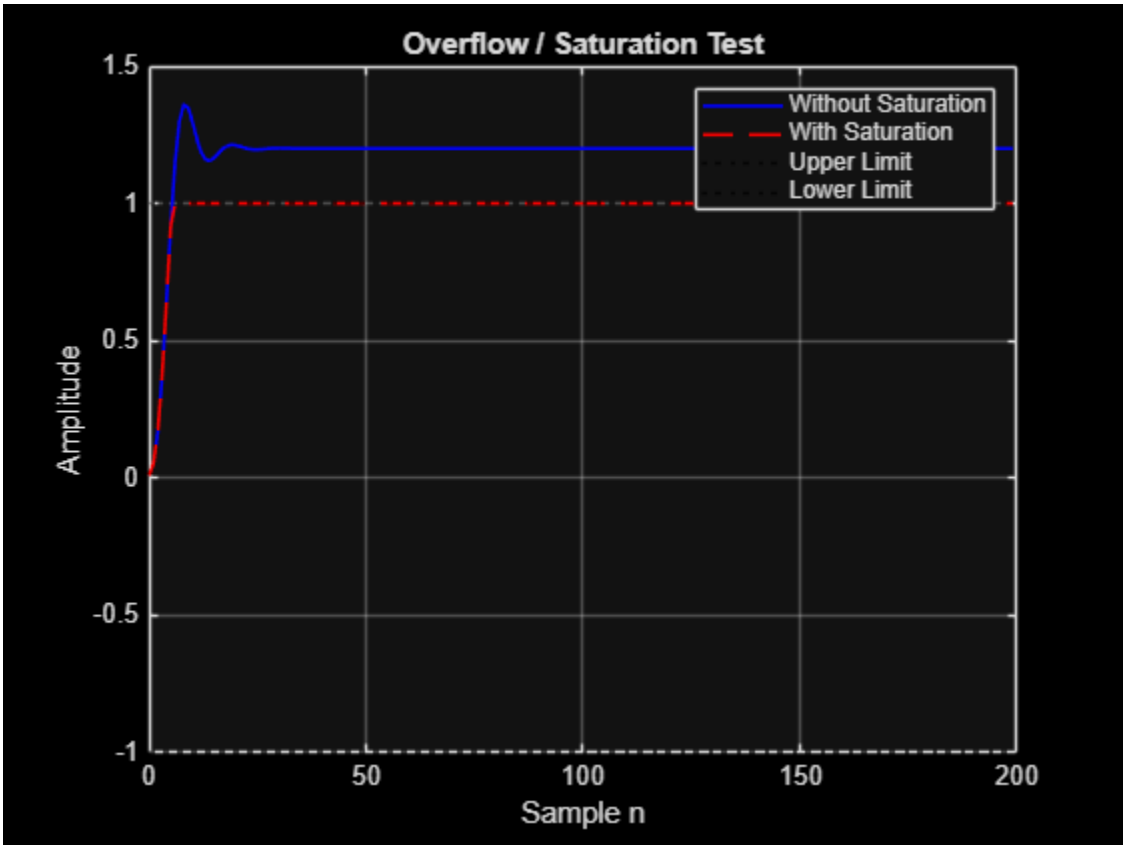
13. GRAPH 5: OVERFLOW / SATURATION TEST

```
figure('Name','Overflow and Saturation');

plot(n,y_over,'b','LineWidth',1.2);
hold on;
plot(n,y_sat,'r--','LineWidth',1.2);

yline(1,'k:');
yline(-1,'k:');

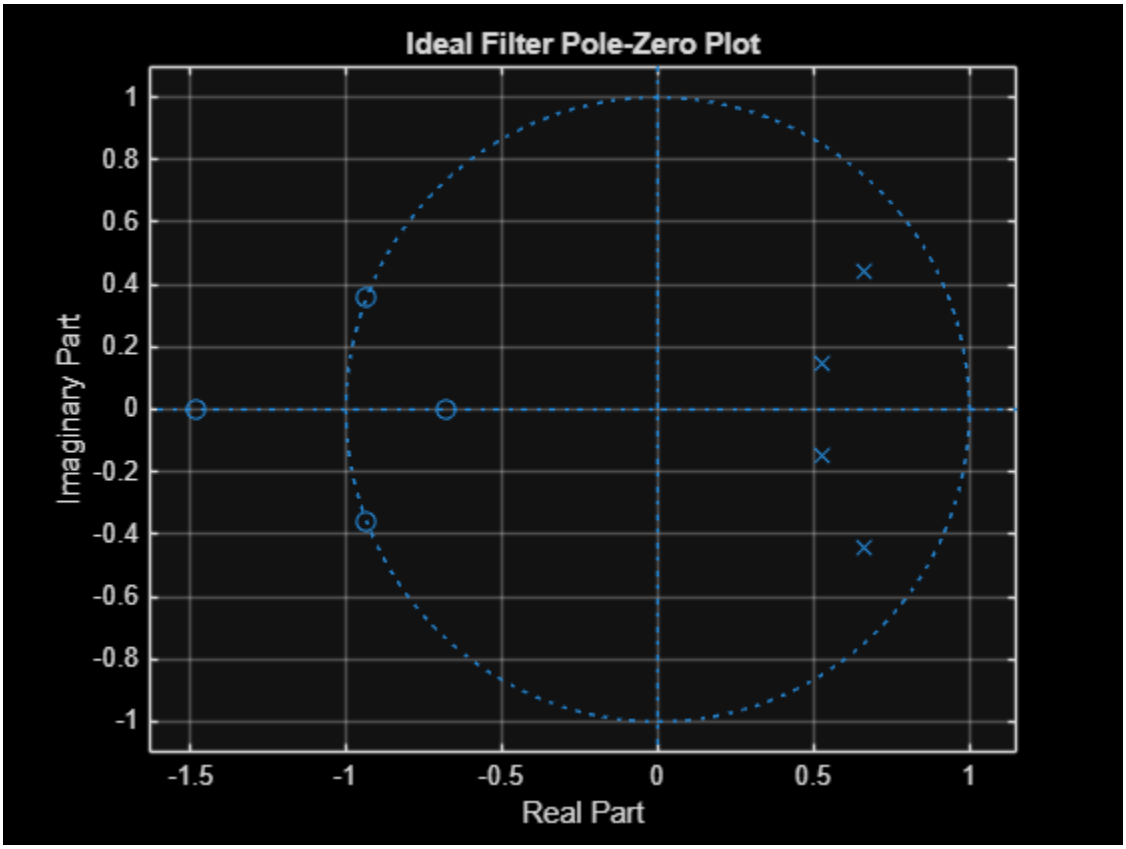
grid on;
xlabel('Sample n');
ylabel('Amplitude');
title('Overflow / Saturation Test');
legend('Without Saturation','With Saturation', ...
       'Upper Limit','Lower Limit');
```



14. GRAPH 6: IDEAL FILTER POLE-ZERO PLOT

```
figure('Name','Ideal Filter Pole-Zero Plot');

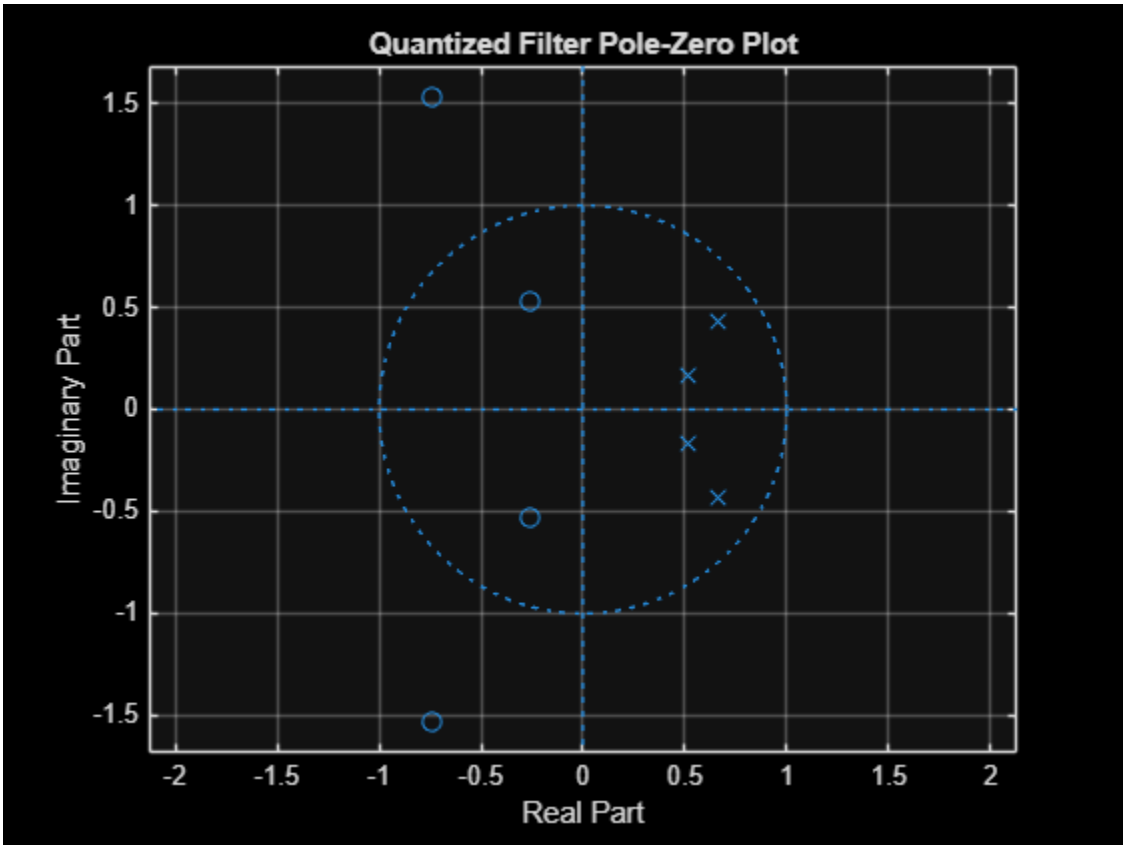
zplane(b,a);
grid on;
title('Ideal Filter Pole-Zero Plot');
```



15. GRAPH 7: QUANTIZED FILTER POLE-ZERO PLOT

```
figure('Name','Quantized Filter Pole-Zero Plot');

zplane(bq,aq);
grid on;
title('Quantized Filter Pole-Zero Plot');
```



16. FINAL RESULT

```
fprintf('\n==========================================\n');
fprintf('              FINAL RESULT\n');
fprintf('==========================================\n');

fprintf('1. Coefficient quantization was performed.\n');
fprintf('2. Product quantization was performed.\n');
fprintf('3. Zero-input behavior was evaluated.\n');
fprintf('4. Overflow and saturation were evaluated.\n');
fprintf('5. Ideal and quantized filter responses were compared.\n');
fprintf('6. Pole-zero locations were plotted for both filters.\n');
fprintf('7. Finite word-length effects were investigated.\n');
```

```
==========================================
              FINAL RESULT
==========================================
1. Coefficient quantization was performed.
2. Product quantization was performed.
3. Zero-input behavior was evaluated.
4. Overflow and saturation were evaluated.
5. Ideal and quantized filter responses were compared.
6. Pole-zero locations were plotted for both filters.
7. Finite word-length effects were investigated.
```

Published with MATLAB® R2026b